UNIT 8

ALGEBRA

INTRODUCTION TO ALGEBRA

An extension of arithmetic in which letters replace the numbers is called Algebra. In other words Algebra is the branch of mathematics that represents problems in the form of mathematical expressions.

For example :
Bisma has 5 toffees in a plate and Hifza has some toffees in her purse.
It can be represent as 5 + x. In this example we have shown the unknown quantity of toffees of Hifza by 'x'.

Example:

In words	In symbols (Algebraic expression)
Sum of 6 and x	$6 + x$
Three times a number y	$3y$
The Product of two numbers x and y	xy
Five is greater than four	$5 > 4$
Double of a number e	$2e$

Sentence

In languages a sentence is set of words making a complete grammatical structure and conveying full meaning.

Example: Represent the following sentences symbolically.
1) Double of a number x is 8.
Answer = $2x = 8$
2) Five is greater than three.
Answer = $5 > 3$

Statement

Sentences that are either true or false are known as statements.

But in some cases we have to decide whether a statement is true or false. Such statement is called **Open statement.**

For example : ▲+ 5 = 12 is an open statement.

Example 1: Think and write the value of which makes the following statements true.

(i) ▲ + 6 = 18

Solution:

▲+ 6 = 18

It will be true if ▲= 12 i.e.

12 + 6 = 18

(ii) ▲– 3 = 7

Solution:

▲- 3 = 7

It will be true if ▲ = 10 i.e.

10 – 3 = 7

Example 2: Replace the unknowns by the numbers to make statement true.

(i) 2 + ▲= 8 **(ii)** 17 -▲= 9

(iii) 10 ÷ ▲ = 5 **(iv)** 6 x ▲ = 42

Solution:

(i) 2 + ▲= 8

It will be true if ▲= 6 i.e. $\boxed{2 + 6 = 8}$

(ii) 17 - ▲= 9

It will be true if ▲= 8 i.e. $\boxed{17 - 8 = 9}$

(iii) 10 ÷▲= 5

It will be true if ▲= 2 i.e. $\boxed{10 \div 2 = 5}$

(iv) 6 x ▲= 42

It will be true if ▲= 7 i.e. $\boxed{6 \times 7 = 42}$

EXERCISE 8.1

Q1. Write the following sentences in symbols.

i. Sum of a number x and two is eight.

ii. Product of a number y and seven is greater than two.

iii. Sum of six and a number z is less than four.

Q2. Which of the following are true or false statements?

(i) 5 + 9 = 18 **(ii)** 8 + 3 = 11 **(iii)** 20 + 3 = 26

(iv) 21 - 8 = 13 **(v)** 18 - 13 = 15 **(vi)** 16 - 10 = 6

Q3 Which of the following are open statements?

(I) x + 3 = 9 **(ii)** 3a + 4 = 8 **(iii)** 9 + 4 = 13

Q4. Write in symbols.

(i) Cost of x books, when price of one book is Rs 18.
(ii) 9b plus 5a
(iii) 3 times x plus 5
(iv) Twice of y minus two third of x
(v) 6 more than p

VARIABLES

Any letter of English alphabet which is used to denote number in algebra is called variable.

For a statement $x + 5 > 2$
for $x = 1$ we get $1 + 5 > 2$ or $6 > 2$
for $x = 2$ we get $2 + 5 > 2$ or $7 > 2$
for $x = 3$ we get $3 + 5 > 2$ or $8 > 2$
and so on
It means the value of number x is not constant, but it is varying.
So, x is called variable.
Know that any numeral, variable or combination of numerals and variables connected by one or more of the symbols '+' and '–' is known as an algebraic expression (e.g., x + 2y)

For example:
(1) Difference of a number x and two. Symbolically we write as $x - 2$
(2) Sum of thrice of a number x and twice of a number y. Symbolically we write as $3x + 2y$

Algebraic expression is the combination of variables, numerals and fundamental operations.

Example:
Moomal has three brothers. Every brother gives x rupees to Moomal. She already has Rs 7.
(i) Write an algebraic expression for the amount she has.
(ii) If each brother gives Rs 10. Find the amount that Moomal has.
(iii) If each brother gives her Rs 15. Find the amount that she has.

Solution:

(i) Three brothers give her $x + x + x = 3x$
Already Moomal has Rs 7
Total amount that Moomal has $3x + 7$

(ii) If $x = 10$ then Moomal has
$3x + 7 = 3 \times 10+7 =$ Rs 37

(iii) If $x = 15$ then Moomal has
$3x + 7 = 3 \times 15+7 =$ Rs 52

Example: Write the number of terms in the following expressions.

(1) $5x + 7y$ **(2)** $6 + y$ **(3)** $3x - 4y - 7$ **(4)** $4\ pq$

Solution:

(1) $5x + 7y$
Number of terms = 2

(2) $6x + 7$
Number of terms = 2

(3) $3x - 4y - 7$
Number of terms = 3

(4) $4\ pq$
Number of terms = 1

Coefficient

In algebraic expression, the symbol or number appearing as multiple of a variable is called its coefficient. e.g. in $4x$, 4 is the coefficient of x.

Variable

A variable is a symbol or letter used to represent an unknown or changeable quantity.

Constant

In algebraic expression number appearing independent of a variable is called constant term.

Example: Write coefficients, constants and variables of the following expressions.

(1) $6z - 8$ **(2)** $2x + 3y + 8$

Solution:

(1) $6z - 8$
Here Coefficient is: 6
Constant is: -8
Variable is : z

(2) $2x + 3y + 8$
Here Coefficients are: 2, 3
Constant is: 8
Variables are: x, y

Like terms can be combined to give a single term.

Addition or subtraction can not be performed with unlike terms.

EXERCISE 8.2

Q1. Write the number of terms of the following expressions.

(i) $3mn$ **(ii)** p **(iii)** $2x + 3y$
(iv) $4y + 7$ **(v)** $4t + 3m + 9$ **(vi)** $7x + 5 + 8$

Q2. Write the variables of the following expressions.

(i) $8x + 3$ **(ii)** $-7x + 1$ **(iii)** $6x + y$ **(iv)** $\frac{2x}{y}$
(v) $2xy$ **(vi)** $5yz$ **(vii)** $3xyz$ **(viii)** $3xy + 9yz$

Q3. Write the constants of the following expressions.

(i) $2x + 2$ **(ii)** $-3 + x$ **(iii)** $xy + 1$
(iv) $xy + yz + \frac{1}{2}$ **(v)** $5x + 9$ **(vi)** $2x + 2y + 10$

Q4. Write the coefficients of variables of the following.

(i) $-7x$ **(ii)** $5y$ **(iii)** $2x + 3y$
(iv) $x + \frac{1}{2}y - \frac{1}{4}z$ **(v)** $5x + 9$ **(vi)** $\frac{1}{4}x + 6y + 1$

Q5. Write like terms in the following.

$2xy$, $4lm$, $-7xz$, $140xy$, $13mn$, xz, $\frac{1}{5}xy$, $-9xyz$, $\frac{1}{4}xy$, $-2p$, $\frac{1}{3}xy$, $-2xy$, $5xy$, xyz, $46p$, $5lm$

ADDITION

Addition of algebraic expressions

Example 1: Add $7x$ and $5x$

Solution:

Writing the terms vertically:

$$\begin{array}{r} 7\,x \\ +\ 5\,x \\ \hline 12\,x \\ \hline \end{array}$$

Writing the terms horizontally:

$7x + 5x = (7 + 5)x = 12x$

Example 2: Find the sum of $3a + 2b + c$ and $8c + 6b + a$

Solution:

Writing the terms vertically:

$$\begin{array}{r} 3a + 2b + c \\ +\quad a + 6b + 8c \\ \hline 4a + 8b + 9c \\ \hline \end{array}$$

Writing the terms horizontally:

Sum

$= (3a + 2b + c) + (a + 6b + 8c)$

$= 3a + 2b + c + a + 6b + 8c$

$= 3a + a + 2b + 6b + c + 8c$

$= 4a + 8b + 9c$

- While adding vertically like terms are arranged in same column and then added.
- While adding horizontally like terms are arranged together and added.

SUBTRACTION

Subtraction of Algebraic Expressions

Example 1: Subtract $6x$ from $9x$.

Solution:

Writing the terms horizontally:

Difference

$= (+9x) - (+6x) = (+9x) + (-6x)$

$= 9x - 6x = 3x$

Writing the terms vertically:

$$\begin{array}{r} 9\,x \\ \overset{+}{-}\ 6\,x \\ \hline 3\,x \\ \hline \end{array}$$

Example 2: Subtract $-6y$ from $10y$.

Solution:

Writing the terms horizontally:

Difference

$= (+10y) - (-6y) = (+10y) + (+6y)$

$= 10y + 6y = 16y$

Writing the terms vertically:

$$\begin{array}{r} 10\,y \\ \underline{\mp\ 6\,y} \\ \underline{16\,y} \end{array}$$

Example 3: Subtract $3x - 2y$ from $8x + 4y$.

Solution:

$(8x + 4y) - (3x - 2y) = 8x + 4y - 3x + 2y$

$= (8x - 3x) + (4y + 2y)$

$= 5x + 6y$

This question can also be solved as under:

$$\begin{array}{r} +8x + 4y \\ \underline{\pm 3x \mp 2y} \\ \underline{5x + 6y} \end{array} \quad \text{(by changing signs)}$$

EXERCISE 8.3

Q1. Add the following expression.

(i) $2x, x, 4x$ **(ii)** $2a, 3a, 6a, a$

(iii) $8mn, 4mn, mn, 6mn$ **(iv)** $2xy, 4xy, xy, 6xy, 3xy$

Q2. Simplify the following:

(i) $2x + 9x$ **(ii)** $a + 2a + 3a$

(iii) $x + 3x + 6x + 10x$ **(iv)** $2st + 3st + 5st + 7st$

Q3. Add the following:

(i) $3a + 2b, a + b, 4a$ **(ii)** $6x + 5y + 7z, 2x + 3y + z, y + 2x$

(iii) $pq + qr + pr, qr + 4pr + 2pq, 3pq + 2pr$

Q4. Subtract.

(i) $5x$ from $8x$ **(ii)** $-3y$ from $9y$

(iii) $2x + 3y$ from $6x + 8y$

(iv) Subtract $-20f + 30g + 40$ from $20 + 10f + 20g$.

Q5. Perform the subtraction.

(i)
$$\begin{array}{r} 8x \\ +5x \\ \hline \\ \hline \end{array}$$

(ii)
$$\begin{array}{r} 12ab \\ -9ab \\ \hline \\ \hline \end{array}$$

(iii)
$$\begin{array}{r} 5z \\ +3z \\ \hline \\ \hline \end{array}$$

(iv)
$$\begin{array}{r} 10xy \\ -17xy \\ \hline \\ \hline \end{array}$$

(v)
$$\begin{array}{r} 2x + 7y \\ -3x + 5y \\ \hline \\ \hline \end{array}$$

(vi)
$$\begin{array}{r} 2x + 15xb + 9y \\ +x + 20xb + 3y \\ \hline \\ \hline \end{array}$$

SIMPLIFICATION

Simplify algebraic expressions grouped with brackets

(i) “____” is called a bar or vinculum.

(ii) "( )" is called a round or curved brackets or parentheses.

(iii) "{ }" is called a curly brackets or braces.

(iv) "[ ]" is called box brackets or square brackets.

Example 1: Simplify the following.

(i) $[5a - \{3b + (6a - 2a + b)\}]$

(ii) $[2a + \{c - a + (a + 2b + c)\}]$

(iii) $xy - [yz - \{zx + xy + (yz - \overline{zx + xy})\}]$

Solutions:

(i) $[5a - \{3b + (6a - 2a + b)\}]$
$= [5a - \{3b + (4a + b)\}]$
$= [5a - \{3b + 4a + b\}]$
$= [5a - 4a - 3b - b]$
$= [5a - 4a - 4b]$
$= [a - 4b] = a - 4b$

(ii) $[2a + \{c - a + (a + 2b + c)\}]$
$= [2a + \{c - a + a + 2b + c\}]$
$= [2a + \{2c + 2b\}]$
$= [2a + 2b + 2c]$
$= 2a + 2b + 2c$

(iii) $xy - [yz - \{zx + xy + (yz - \overline{zx + xy})\}]$
$= xy - [yz - \{zx + xy + yz - zx - xy\}]$
$= xy - [yz - \{\cancel{zx} + \cancel{xy} - yz - \cancel{zx} + \cancel{xy}\}]$
$= xy - [yz - yz] = xy - [\cancel{yz} - \cancel{yz}] = xy - 0 = xy$

Example 2: Simplify the following expression.
$25 - [-7a - \{-6a + (3 - 5 - 6a)\}]$

Solution:
$= 25 - [-7a - \{-6a + (3 - 5 - 6a)\}]$
$= 25 - [-7a - \{-6a + (-2 - 6a)\}]$
$= 25 - [-7a - \{-6a - 2 - 6a\}]$
$= 25 - [-7a - \{-12a - 2\}]$
$= 25 - [-7a + 12a + 2]$
$= 25 - [5a + 2] = 25 - 5a - 2 = 23 - 5a$

Example: If $a = 2$, $b = -3$ and $c = -4$, then evaluate the following.

(i) $ab + bc + ca$ **(ii)** $\dfrac{bc\,(b - c)}{a}$

Solution:

(i) $ab + bc + ca$
$= 2(-3) + (-3)(-4) + (-4)(2)$
$= -6 + 12 + -8$
$= -14 + 12$
$= -2$

(ii) $\dfrac{bc\,(b - c)}{a}$

$= \dfrac{(-3)(-4)[(-3) - (-4)]}{2}$

$= \dfrac{12[-3 + 4]}{2} = \dfrac{\cancel{12}^{6}[1]}{\cancel{2}_{1}} = 6$

EXERCISE 8.4

Q1. Simplify the following expressions.

a) $[x + x + (y + y + 2x)]$ **b)** $[5a - \{3a + (3b - 6b + 4a)\}]$
c) $2a - \{5b - 3(2a + b - 3a)\}$
d) $5m - [2 - (3 + 11m + 3n + 8m)]$

Q2. If $a = -3$, $b = 5$ and $c = -2$, evaluate the following:

(i) $a + b + c$ **(ii)** $2a - 3b + c$

(iii) $\dfrac{a - bc}{c}$

(iv) $\dfrac{2a + b + c}{abc}$

(v) $\dfrac{a + b}{c} + \dfrac{b + c}{a}$

Q3. (i) If x = 2, find the values of 3x and – 5x
(ii) If y = – 3, find the values of the following expressions.

(a) $-3y$

(b) $5y + 7$